Yogesh Gandhi

Computational Mechanics · Finite Element Simulation · Design Optimization

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SUMMARY

I build simulation tools for engineering design, and I like taking an idea all the way from the math to a real, manufactured part. My background spans computational mechanics broadly — from optimization and control to finite-element simulation and structural design optimization — first in academic research, then applied to real production components, and now in industry. Today I'm a Senior Researcher at Braid Technologies in Tokyo, building the finite-element engine behind automated engineering design. The thread through all of it is the same: optimization, and making methods actually work on real problems.

SKILLS

- **Programming:** Python · MATLAB · Fortran
- **Simulation / CAE:** LS-DYNA · Kratos Multiphysics · MFEM · Abaqus · ANSYS · GMSH · SolidWorks
- **Methods:** Topology & shape optimization · FEA (linear/nonlinear, implicit/explicit) · isogeometric analysis (IGA) · adjoint & stress-constrained sensitivity · differentiable simulation (automatic differentiation) · composites & laminate theory · additive manufacturing · multiscale modelling · failure & damage
- **Tools:** Linux · Git · LaTeX

EXPERIENCE

Senior Researcher — Braid Technologies, Tokyo, Japan *Aug 2025 – Present*

- Build the finite-element engine behind automated engineering design — software that optimizes a part's shape and topology on its own, replacing slow manual design iteration for industrial clients and cutting design-cycle time by roughly 10×.
- Develop the element library: solid elements (linear and quadratic tetrahedra and hexahedra) and shell elements (corotational shells for large-rotation nonlinearity, and isogeometric NURBS Kirchhoff–Love shells that run analysis directly on CAD geometry).
- Build the nonlinear (static and dynamic), modal, and contact/constraint solving capability, and integrate commercial and open-source solvers (LS-DYNA, Kratos, MFEM) behind one consistent interface.
- Implement gradient-based topology and shape optimization with automatic-differentiation sensitivities — end-to-end design gradients that replace slow finite-difference loops, including stress-constrained design.

Postdoctoral Researcher — University of Bologna, Forlì, Italy *Jan 2024 – Jun 2025*

- Developed topology- and stress-based design-optimization methods for additively manufactured structural parts, taking them from formulation to production hardware.
- Built and applied a stress-driven design framework for 3D-printed structural components of a Ferrari 296GTB mechanism, achieving roughly 18% weight reduction and 25% higher stiffness.
- Co-led the structural design, analysis, and additive manufacturing of parts for an award-winning solar vehicle (winner, Sasol 2024 Cruiser Challenge).

Research Associate — University of Parma, Parma, Italy *Apr 2018 – Feb 2020*

- Built constitutive and design-optimization models for advanced materials — thermo-mechanical (shape-memory-alloy) modelling via custom material subroutines, and optimization of fiber layouts for tailored structural response.
- Modelled fracture and damage in composite structures, including delamination growth.

Selected research collaborations (during PhD)

- TU Delft (2021–2024) — CAD-based topology optimization of continuous-fiber laminates, with Prof. Alejandro Aragón and Prof. Julián Norato.
- Politecnico di Torino (2023–2024) — topology optimization using the Carrera Unified Formulation.
- University of Padova (2023–2025) — 9T Labs process optimization, material characterization, and model validation, with Prof. Paolo Carraro.

EDUCATION

PhD, Aerospace Engineering — University of Bologna, Forlì, Italy *2020 – 2024 (awarded Jun 2024)*

Thesis: a unified geometry-projection topology-optimization approach for additively manufactured, variable-stiffness, continuous fiber-reinforced composite laminates.

MSc, Aerospace Engineering — University of Bologna, Forlì, Italy *2015 – 2017*

Master's thesis (Coventry University, UK): optimal path-planning (A, D*) and nonlinear control for a mobile robot.*

BE, Mechanical Engineering — PES Institute of Technology, Bengaluru, India *2010 – 2014*

SELECTED PUBLICATIONS

126 citations · h-index 5 · i10-index 4 (Google Scholar, 2026)

- **Y. Gandhi**, A. Aragón, J. Norato, G. Minak. *A geometry projection method for the topology optimization of additively manufactured variable-stiffness composite laminates.* Computer Methods in Applied Mechanics and Engineering, 2025. doi:10.1016/j.cma.2024.117663

Two further journal articles (*Applied Sciences*, 2022; *Materials*, 2019), conference proceedings, and six international talks — full list on request or at gandhiyudheer.github.io.

TEACHING, MENTORING & SERVICE

- **Teaching & mentoring:** Topology-optimization theory + MATLAB (MSc, University of Bologna, 2023–2024) and composite failure in Abaqus (Dallara Academy, 2019); co-supervised a PhD candidate and an MSc student.
- **Service & award:** Peer reviewer for *Composite Structures* and *Additive Manufacturing*; winner, Sasol 2024 Cruiser Challenge (Emilia V).

LANGUAGES

English (C1) · Italian (B2) · Japanese (beginner, learning)